# Mark Scheme

Q1.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (a) | $2 < x < 6$ | B1 | 1.1b |
| | | (1) | |
| (b) | States either $k > 8$ or $k < 0$ | M1 | 3.1a |
| | States e.g. $\{k : k > 8\} \cup \{k : k < 0\}$ | A1 | 2.5 |
| | | (2) | |
| (c) | **Please see notes for alternatives** | | |
| | States $y = ax(x-6)^2$ or $f(x) = ax(x-6)^2$ | M1 | 1.1b |
| | Substitutes $(2, 8)$ into $y = ax(x-6)^2$ and attempts to find $a$ | dM1 | 3.1a |
| | $y = \frac{1}{4}x(x-6)^2$ or $f(x) = \frac{1}{4}x(x-6)^2$ o.e | A1 | 2.1 |
| | | (3) | |
| | | **(6 marks)** | |
| **Notes: Watch for answers written by the question. If they are beside the question and in the answer space, the one in the answer space takes precedence** | | | |

(a)

B1: Deduces $2 < x < 6$ o.e. such as $x > 2, x < 6$ $\;x > 2$ and $x < 6$ $\;\{x : x > 2\} \cap \{x : x < 6\}$ $\;x \in (2, 6)$

Condone attempts in which set notation is incorrectly attempted but correct values can be seen or implied E.g. $\{x > 2\} \cap \{x < 6\}$ $\{x > 2, x < 6\}$. Allow just the open interval $(2, 6)$

Do not allow for incorrect inequalities such as e.g. $x > 2$ or $x < 6$, $\{x : x > 2\} \cup \{x : x < 6\}$ $x \in [2, 6]$

(b)

M1: Establishes a correct method by finding one of the (correct) inequalities

States either $k > 8$ (condone $k \geqslant 8$) or $k < 0$ (condone $k \leqslant 0$)

Condone for this mark $y \leftrightarrow k$ or $f(x) \leftrightarrow k$ and $8 < k < 0$

A1: Fully correct solution in the form $\{k : k > 8\} \cup \{k : k < 0\}$ or $\{k | k > 8\} \cup \{k | k < 0\}$ either way around but condone $\{k < 0\} \cup \{k > 8\}$, $\{k : k < 0 \cup k > 8\}$, $\{k < 0 \cup k > 8\}$. It is not necessary to mention $\mathbb{R}$, e.g. $\{k : k \in \mathbb{R}, k > 8\} \cup \{k : k \in \mathbb{R}, k < 0\}$ Look for $\{\ \}$ and $\cup$

Do not allow solutions not in set notation such as $k < 0$ or $k > 8$.

(c)

M1: Realises that the equation of $C$ is of the form $y = ax(x-6)^2$. Condone with $a = 1$ for this mark.

So award for sight of $ax(x-6)^2$ even with $a = 1$

dM1: Substitutes $(2, 8)$ into the form $y = ax(x-6)^2$ and attempts to find the value for $a$.

It is dependent upon having an equation, which the ($y = ..$) may be implied, of the correct form.

A1: Uses all of the information to form a correct **equation** for $C$ $y = \frac{1}{4}x(x-6)^2$ o.e.

ISW after a correct answer. Condone $f(x) = \frac{1}{4}x(x-6)^2$ but not $C = \frac{1}{4}x(x-6)^2$.

Allow this to be written down for all 3 marks

**Alternative I part (c):**
**Using the form $y = ax^3 + bx^2 + cx$ and setting up then solving simultaneous equations.**
**There are various versions of this but can be marked similarly**

M1: Realises that the equation of $C$ is of the form $y = ax^3 + bx^2 + cx$ and forms two equations in $a$, $b$ and $c$. Condone with $a$ =1 for this mark.

Note that the form $y = ax^3 + bx^2 + cx + d$ is M0 until $d$ is set equal to 0.

There are four equations that could be formed, only two are necessary for this mark.

Condone slips

Using (6, 0) $\Rightarrow 216a+36b+6c=0$

Using (2, 8) $\Rightarrow 8a+4b+2c=8$

Using $\frac{dy}{dx}=0$ at $x=2 \Rightarrow 12a+4b+c=0$

Using $\frac{dy}{dx}=0$ at $x=6 \Rightarrow 108a+12b+c=0$

dM1: Forms and solves three different equations, one of which must be using (2, 8) to find values for $a$, $b$ and $c$. A calculator can be used to solve the equations

A1: Uses all of the information to form a correct equation for $C$ $y=\frac{1}{4}x^3-3x^2+9x$ o.e.

ISW after a correct answer. Condone $f(x)=\frac{1}{4}x^3-3x^2+9x$

**Alternative II part (c)**
**Using the gradient and integrating**

M1: Realises that the gradient of $C$ is zero at 2 and 6 so sets $f'(x)=k(x-2)(x-6)$ oe **and** attempts to integrate. Condone with $k=1$

dM1: Substitutes $x=2, y=8$ into $f(x)=k\left(...x^3+...x+...\right)$ and finds a value for $k$

A1: Uses all of the information to form a correct equation for $C$ $y=\frac{3}{4}\left(\frac{1}{3}x^3-4x^2+12x\right)$ o.e.

ISW after a correct answer. Condone $f(x)=\frac{3}{4}\left(\frac{1}{3}x^3-4x^2+12x\right)$

EXAM PAPERS PRACTICE

Q2.

| Part | Working or answer an examiner might expect to see | Mark | Notes |
|---|---|---|---|
| (a) | $\frac{dy}{dx} = \frac{(x+1)^2 \times (10x+10) - (5x^2+10x) \times 2(x+1)}{(x+1)^4}$ | M1 | This mark is given for an attempt to differentiate the expression for $y$ |
| | | A1 | This mark is given for correctly differentiating the expression for $y$ |
| | $\frac{dy}{dx} = \frac{(x+1) \times (10x+10) - (5x^2+10x) \times 2}{(x+1)^3}$ | M1 | This mark sis given for cancelling the expression through by $(x + 1)$ |
| | $\frac{dy}{dx} = \frac{10}{(x+1)^3}$ | A1 | This mark is given for a fully correct expression for $\frac{dy}{dx}$ |
| (b) | If $A > 0$ and $n = 1, 3$ then $x < -1$ | B1 | This mark is given for deducing that $\frac{dy}{dx} < 0 \Rightarrow x < -1$. |

Q3.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (i) | $x^2 - 6x + 10 = (x-3)^2 + 1$ | M1 | 2.1 |
| | Deduces "always true" as $(x-3)^2 \geqslant 0 \Rightarrow (x-3)^2 + 1 \geqslant 1$ and so is always positive | A1 | 2.2a |
| | | (2) | |
| (ii) | For an explanation that it need not (always) be true This could be if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$ | M1 | 2.3 |
| | States 'sometimes' and explains if $a > 0$ then $ax > b \Rightarrow x > \frac{b}{a}$ if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$ | A1 | 2.4 |
| | | (2) | |
| (iii) | Difference $= (n+1)^2 - n^2 = 2n+1$ | M1 | 3.1a |
| | Deduces "Always true" as $2n+1 = (\text{even} + 1) = \text{odd}$ | A1 | 2.2a |
| | | (2) | |
| | | | **(6 marks)** |

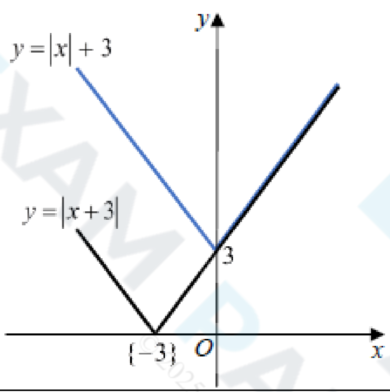
EXAM PAPERS PRACTICE

| **Notes:** |
| --- |
| **(i)** |
| **M1:** Attempts to complete the square or any other valid reason. Allow for a graph of $y = x^2 - 6x + 10$ or an attempt to find the minimum by differentiation |
| **A1:** States always true with a valid reason for their method |
| **(ii)** |
| **M1:** For an explanation that it need not be true (sometimes). This could be if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$ or simply $-3x > 6 \Rightarrow x < -2$ |
| **A1:** Correct statement (sometimes true) and explanation |
| **(iii)** |
| **M1:** Sets up the proof algebraically. For example by attempting $(n+1)^2 - n^2 = 2n+1$ or $m^2 - n^2 = (m-n)(m+n)$ with $m = n+1$ |
| **A1:** States always true with reason and proof<br>Accept a proof written in words. For example<br>If integers are consecutive, one is odd and one is even<br>When squared $\text{odd} \times \text{odd} = \text{odd}$ and $\text{even} \times \text{even} = \text{even}$<br>The difference between odd and even is always odd, hence always true<br>Score M1 for two of these lines and A1 for a good proof with all three lines or equivalent. |

Q4.

| Part | Working or answer an examiner might expect to see | Mark | Notes |
|---|---|---|---|
| (a) | $g(0) = 5$ | M1 | This mark is given for a method to find $g(0)$ |
| | $gg(0) = g(5) = 13$ | A1 | This mark is given for a correct value for $gg(0)$ |
| (b) | $(x-2)^2 + 1 > 28$<br>$(x-2)^2 > 27$<br>$x - 2 > 3\sqrt{3}$ | M1 | This mark is given for a method to solve $g(x) > 28$ when $x \leq 2$ |
| | $x < 2 - 3\sqrt{3}$ | A1 | |
| | $4x - 7 > 28$<br>$4x > 35$<br>$x > \frac{35}{4}$ | M1 | This mark is given for a solving $g(x) > 28$ when $x > 2$ |
| | $x < 2 - 3\sqrt{3}$ and $x > \frac{35}{4}$ | A1 | This mark is given for a correct range of values of $x$ for which $g(x) > 28$ stated |
| (c) | $h^{-1}$ exists since h is a one-to-one function; $g^{-1}$ does not exists since g is a many-to-one function | B1 | This mark is given for a valid explanation |
| (d) | $h^{-1}(x) = 2 - \sqrt{(x-1)}$ | B1 | This mark is given for finding an expression for $h^{-1}(x)$ |
| | $2 \pm \sqrt{(x-1)} = -\frac{1}{2}$ | M1 | This mark is given for a method to rearrange to find a value for $x$ |
| | $x = 7.25$ | A1 | This mark is given for a correct value of $x$ |
| | | | **(Total 10 marks)** |

Q5.

| Question | Scheme | | Marks | AOs |
|---|---|---|---|---|
| | Statement: "If $m$ and $n$ are irrational numbers, where $m \neq n$, then $mn$ is also irrational." | | | |
| **(a)** | E.g. $m = \sqrt{3},\ n = \sqrt{12}$ | | M1 | 1.1b |
| | $\{mn =\}\ \left(\sqrt{3}\right)\left(\sqrt{12}\right) = 6$<br>$\Rightarrow$ statement untrue **or** 6 is not irrational **or** 6 is rational | | A1 | 2.4 |
| | | | **(2)** | |
| **(b)(i), (ii) Way 1** |  | V shaped graph {reasonably} symmetrical about the $y$-axis with vertical interpret $(0, 3)$ or 3 stated or marked on the positive $y$-axis | B1 | 1.1b |
| | | Superimposes the graph of $y = \lvert x+3\rvert$ on top of the graph of $y = \lvert x\rvert + 3$ | M1 | 3.1a |
| | the graph of $y = \lvert x\rvert + 3$ is either the same or above the graph of $y = \lvert x+3\rvert$ {for corresponding values of $x$}<br>**or** when $x \geq 0$, both graphs are equal (or the same)<br>when $x < 0$, the graph of $y = \lvert x\rvert + 3$ is above the graph of $y = \lvert x+3\rvert$ | | A1 | 2.4 |
| | | | **(3)** | |
| **(b)(ii) Way 2** | **Reason 1**<br>When $x \geq 0$, $\lvert x\rvert + 3 = \lvert x+3\rvert$ | Any one of Reason 1 or Reason 2 | M1 | 3.1a |
| | **Reason 2**<br>When $x < 0$, $\lvert x\rvert + 3 > \lvert x+3\rvert$ | Both Reason 1 and Reason 2 | A1 | 2.4 |
| | | | | **(5 marks)** |

| Notes for Question | |
|---|---|
| **(a)** | |
| **M1:** | States or uses any pair of ***different*** numbers that will disprove the statement. E.g. $\sqrt{3},\ \sqrt{12}$ ; $\sqrt{2},\ \sqrt{8}$ ; $\sqrt{5},\ -\sqrt{5}$; $\frac{1}{\pi},\ 2\pi$; $3\text{e},\ \frac{4}{5\text{e}}$; |
| **A1:** | Uses correct reasoning to disprove the given statement, with a correct conclusion |
| **Note:** | Writing $(3\text{e})\left(\frac{4}{5\text{e}}\right)=\frac{12}{5} \Rightarrow$ untrue is sufficient for M1A1 |
| **(b)(i)** | |
| **B1:** | See scheme |
| **(b)(ii)** | |
| **M1:** | For constructing a method of comparing $\lvert x\rvert+3$ with $\lvert x+3\rvert$. See scheme. |
| **A1:** | Explains fully why $\lvert x\rvert+3 \geq \lvert x+3\rvert$. See scheme. |
| **Note:** | Do not allow either $x>0,\ \lvert x\rvert+3 \geq \lvert x+3\rvert$ or $x \geq 0,\ \lvert x\rvert+3 \geq \lvert x+3\rvert$ as a valid reason |
| **Note** | $x=0$ (or where necessary $x=-3$) need to be considered in their solutions for A1 |
| **Note:** | Do not allow an incorrect statement such as $x \leq 0,\ \lvert x\rvert+3 > \lvert x+3\rvert$ for A1 |
| **(b)(ii)** | |
| **Note:** | Allow M1A1 for $x>0,\ \lvert x\rvert+3 = \lvert x+3\rvert$ and for $x \leq 0,\ \lvert x\rvert+3 \geq \lvert x+3\rvert \geqslant$ |
| **Note:** | Allow M1 for any of<br>• $x$ is positive, $\lvert x\rvert+3=\lvert x+3\rvert$<br>• $x$ is negative, $\lvert x\rvert+3>\lvert x+3\rvert$<br>• $x>0,\ \lvert x\rvert+3=\lvert x+3\rvert$<br>• $x\leq 0,\ \lvert x\rvert+3\geq\lvert x+3\rvert$<br>• $x>0,\ \lvert x\rvert+3$ and $\lvert x+3\rvert$ are equal<br>• $x\geq 0,\ \lvert x\rvert+3$ and $\lvert x+3\rvert$ are equal<br>• when $x\geq 0$, both graphs are equal<br>• for positive values $\lvert x\rvert+3$ and $\lvert x+3\rvert$ are the same<br>Condone for M1<br>• $x\leq 0,\ \lvert x\rvert+3>\lvert x+3\rvert$<br>• $x<0,\ \lvert x\rvert+3\geq\lvert x+3\rvert$ |

| | | | |
|---|---|---|---|
| **(b)(ii) Way 3** | • For $x>0,\ \lvert x\rvert+3=\lvert x+3\rvert$<br>• For $-3<x<0$, as $\lvert x\rvert+3>3$ and $\{0<\}\ \lvert x+3\rvert<3$, then $\lvert x\rvert+3>\lvert x+3\rvert$ | M1 | 3.1a |
| | • For $x\leq -3$, as $\lvert x\rvert+3=-x+3$ and $\lvert x+3\rvert=-x-3$, then $\lvert x\rvert+3>\lvert x+3\rvert$ | A1 | 2.4 |

Q6.

| Part | Working or answer an examiner might expect to see | Mark | Notes |
|---|---|---|---|
| (a) | $\log a - \log b = \log \frac{a}{b}$ | B1 | This mark is given for restating the log equation using $\log \frac{a}{b}$ |
| | $a - b = \frac{a}{b}$<br>$ab - b^2 = a$<br>$ab - a = b^2$ | M1 | This mark is given for rearranging so that terms in a are on one side of the equation |
| | $a(b - 1) = b^2$<br>$a = \frac{b^2}{(b-1)}$ | A1 | This mark is for rearranging to show the result required |
| (b) | $b \neq 1$ | B1 | This mark is given for deducing that $b \neq 1$ |
| | Since $a > 0$, $\frac{b^2}{(b-1)} > 0$<br>$b > 1$ since $b^2$ is positive | B1 | This mark is given for stating that $b > 1$ and explaining the reason for the restriction |

Q7.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| **(a)** | Attempts $(x-2)^2 + (y+5)^2 = ....$ | M1 | 1.1b |
| | Centre $(2, -5)$ | A1 | 1.1b |
| | | **(2)** | |
| **(b)** | Sets $k + 2^2 + 5^2 > 0$ | M1 | 2.2a |
| | $\Rightarrow k > -29$ | A1ft | 1.1b |
| | | **(2)** | |
| | | | **(4 marks)** |

**Notes:**

**(a)**

**M1:** Attempts to complete the square so allow $(x-2)^2 + (y+5)^2 = ....$

**A1:** States the centre is at $(2, -5)$. Also allow written separately $x = 2, y = -5$

$(2, -5)$ implies both marks

**(b)**

**M1:** Deduces that the right hand side of their $(x \pm ...)^2 + (y \pm ...)^2 = ...$ is $> 0$ or $\geqslant 0$

**A1ft:** $k > -29$ Also allow $k \geqslant -29$ Follow through on their rhs of $(x \pm ...)^2 + (y \pm ...)^2 = ...$

Q8.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (a) | $f(x) \geqslant 5$ | B1 | 1.1b |
| | | (1) | |
| (b) | Uses $-2(3-x)+5=\frac{1}{2}x+30$ | M1 | 3.1a |
| | Attempts to solve by multiplying out bracket, collect terms etc $\frac{3}{2}x=31$ | M1 | 1.1b |
| | $x=\frac{62}{3}$ only | A1 | 1.1b |
| | | (3) | |
| (c) | Makes the connection that there must be two intersections. Implied by either end point $k>5$ or $k \leqslant 11$ | M1 | 2.2a |
| | $\{k : k \in \mathbb{R}, 5 < k \leqslant 11\}$ | A1 | 2.5 |
| | | (2) | |
| | | | **(6 marks)** |

**Notes:**

**(a)**

**B1:** $f(x) \geqslant 5$ Also allow $f(x) \in [5, \infty)$

**(b)**

**M1:** Deduces that the solution to $f(x)=\frac{1}{2}x+30$ can be found by solving $-2(3-x)+5=\frac{1}{2}x+30$

**M1:** Correct method used to solve their equation. Multiplies out bracket/ collects like terms

**A1:** $x=\frac{62}{3}$ only. Do not allow 20.6

**(c)**

**M1:** Deduces that two distinct roots occurs when $y=k$ intersects $y=f(x)$ in two places. This may be implied by the sight of either end point. Score for sight of either $k>5$ or $k \leqslant 11$

**A1:** Correct solution only $\{k : k \in \mathbb{R}, 5 < k \leqslant 11\}$

Q9.

| Part | Working or answer an examiner might expect to see | Mark | Notes |
|---|---|---|---|
| (a) | $y = a + kx$, where $a$ and $k$ are constants | B1 | This mark is given for stating a correct general equation |
| (b) | $500 = 800 \times 2 - (a + 800k)$<br>$-80 = 300 \times 2 - (a + 300k)$ | M1 | This mark is given for modelling the profit on the two days when bars of soap are sold for £2 |
| | $a + 800k = 1100$<br>$a + 300k = 680$ | M1 | This mark is given for forming a pair of simultaneous equations to find values for $a$ and $k$ |
| | $500k = 420 \Rightarrow k = \frac{420}{500} = 0.84$<br>$a + (800 \times 0.84) = 1100$<br>$a = 1100 - 672 = 428$<br>Thus $y = 0.84x + 428$ | A1 | This mark is given for finding the values of $a$ and $k$ to show $y = 0.84x + 428$ as required |
| (c) | 0.84 represents the cost of making one extra bar of soap in £s (i.e. 84p) | B1 | This mark is given for a valid interpretation of the significance of 0.84 |
| (d) | For $n$ bars of soap<br>$2n - (428 + 0.84n) > 0$ | M1 | This mark is given for a method to find the number of bars of soap to be made |
| | $1.16n - 428 > 0$<br>$n - \frac{428}{1.16} > 0$<br>$n = 369$ bars of soap | A1 | This mark is given for correctly finding the number of bars of soap to be made |
| | | | **(Total 7 marks)** |

Q10.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| | $15 - 2^{x+1} = 3 \times 2^x$ | B1 | 1.1b |
| | $\Rightarrow 15 - 2 \times 2^x = 3 \times 2^x \Rightarrow 2^x = 3$<br>or e.g.<br>$\Rightarrow \frac{15}{2^x} - 2 = 3 \Rightarrow 2^x = 3$ | M1 | 1.1b |
| | $2^x = 3 \Rightarrow x = ...$ | dM1 | 1.1b |
| | $x = \log_2 3$ | A1cso | 1.1b |
| | | (4) | |
| | Alternative | | |
| | $y = 3 \times 2^x \Rightarrow 2^x = \frac{y}{3} \Rightarrow y = 15 - 2 \times \frac{y}{3}$ | B1 | 1.1b |
| | $3y + 2y = 45 \Rightarrow y = 9 \Rightarrow 3 \times 2^x = 9 \Rightarrow 2^x = 3$ | M1 | 1.1b |
| | $2^x = 3 \Rightarrow x = ...$ | dM1 | 1.1b |
| | $x = \log_2 3$ | A1cso | 1.1b |
| | | | **(4 marks)** |

## Notes:

**B1:** Combines the equations to reach $15-2^{x+1}=3\times2^x$ or equivalent e.g. $15-2^{x+1}-3\times2^x=0$

**M1:** Uses $2^{x+1}=2\times2^x$ oe e.g. $\frac{2^{x+1}}{2^x}=2$ to obtain an equation in $2^x$ and attempts to make $2^x$ the subject.

See scheme but e.g. $y=2^x \Rightarrow 3\times2^x=15-2^{x+1} \Rightarrow 3y=15-2y \Rightarrow y=\ldots$ is also possible

**dM1:** Uses logs correctly and proceeds to a value for $x$ from an equation of the form $2^x=k$ where $k>1$

e.g. $2^x=k \Rightarrow x=\log_2 k$

or $2^x=k \Rightarrow \log 2^x=\log k \Rightarrow x\log 2=\log k \Rightarrow x=\ldots$

or $2^x=k \Rightarrow \ln 2^x=\ln k \Rightarrow x\ln 2=\ln k \Rightarrow x=\ldots$

**Depends on the first method mark**

**This may be implied if they go straight to decimals e.g. $2^x=3$ so $x=1.584..$ but you may need to check**

**A1cso:** $x=\log_2 3$ or $\frac{\log 3}{\log 2}$ or $\frac{\ln 3}{\ln 2}$

Ignore any attempts to find the $y$-coordinate

**Alternative**

**B1:** Correct equation in $y$

**M1:** Solves their equation in $y$ and attempts to make $2^x$ the subject.

**dM1:** Uses logs correctly and proceeds to a value for $x$ from an equation of the form $2^x=k$ where $k>1$

e.g. $2^x=k \Rightarrow x=\log_2 k$

or $2^x=k \Rightarrow \log 2^x=\log k \Rightarrow x\log 2=\log k \Rightarrow x=\ldots$

or $2^x=k \Rightarrow \ln 2^x=\ln k \Rightarrow x\ln 2=\ln k \Rightarrow x=\ldots$

**Depends on the first method mark**

**This may be implied if they go straight to decimals e.g. $2^x=3$ so $x=1.584..$ but you may need to check**

**A1cso:** $x=\log_2 3$ or $\frac{\log 3}{\log 2}$ or $\frac{\ln 3}{\ln 2}$

Ignore any attempts to find the $y$-coordinate

Q11.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (a) | $C$ is $(x-r)^2+(y-r)^2=r^2$ or $x^2+y^2-2rx-2ry+r^2=0$ | B1 | 2.2a |
| | $y=12-2x,\ x^2+y^2-2rx-2ry+r^2=0$ $\Rightarrow x^2+(12-2x)^2-2rx-2r(12-2x)+r^2=0$ or $y=12-2x,\ (x-r)^2+(y-r)^2=r^2$ $\Rightarrow(x-r)^2+(12-2x-r)^2=r^2$ | M1 | 1.1b |
| | $x^2+144-48x+4x^2-2rx-24r+4rx+r^2=0$ $\Rightarrow 5x^2+(2r-48)x+(r^2-24r+144)=0$ * | A1* | 2.1 |
| | | (3) | |
| (b) | $b^2-4ac=0 \Rightarrow (2r-48)^2-4\times5\times(r^2-24r+144)=0$ | M1 | 3.1a |
| | $r^2-18r+36=0$ or any multiple of this equation | A1 | 1.1b |
| | $\Rightarrow(r-9)^2-81+36=0 \Rightarrow r=\ldots$ | dM1 | 1.1b |
| | $r=9\pm3\sqrt{5}$ | A1 | 1.1b |
| | | (4) | |
| | | | (7 marks) |

**Notes:**

**(a)**

**B1:** Deduces the correct equation of the circle

**M1:** Attempts to form an equation with terms of the form $x^2$, $x$, $r^2$, and $xr$ **only** using $y = 12 \pm 2x$ and their circle equation which must be of an appropriate form. I.e. includes or implies an $x^2$, $y^2$, $r^2$ such as $x^2 + y^2 = r^2$

If their circle equation starts off as e.g. $(x \pm a)^2 + (y \pm b)^2 = r^2$ then the B mark and the M mark can be awarded when the "$a$" and "$b$" are replaced by $r$ or $-r$ as appropriate for their circle equation.

**A1***: Uses correct and accurate algebra leading to the given solution.

**(b)**

**M1**: Attempts to use $b^2 - 4ac...0$ o.e. with $a = 5, b = 2r - 48, c = r^2 - 24r + 144$ and where ... is "=" or any inequality

Allow minor slips when copying the $a$, $b$ and $c$ provided it does not make the work easier and allow **their** $a$, $b$ and $c$ if they are similar expressions.

FYI $(2r-48)^2 - 4\times5\times(r^2 - 24r + 144) = 4r^2 - 192r + 2304 - 20r^2 + 480r - 2880 = -16r^2 + 288r - 576$

**A1:** Correct quadratic **equation** in $r$ (or inequality). Terms need not be all one side but must be collected.

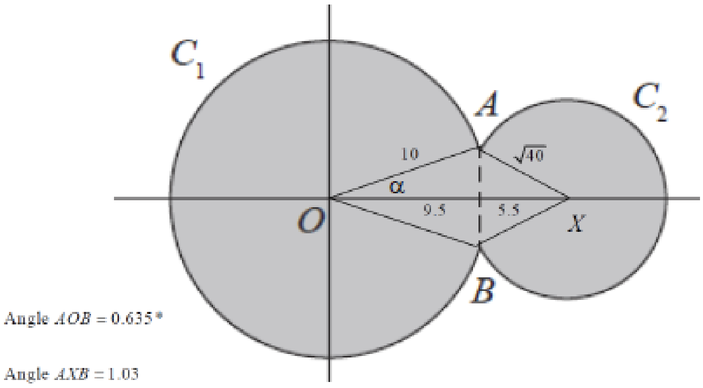
E.g. allow $r^2 - 18r = -36$ and allow any multiple of this equation (or inequality).

**dM1:** Correct attempt to solve their 3TQ in $r$. Dependent upon previous M

**A1:** Careful and accurate work leading to both answers in the required form (must be simplified surds)

Q12.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (a) | Solves $x^2 + y^2 = 100$ and $(x-15)^2 + y^2 = 40$ simultaneously to find $x$ or $y$ E.g. $(x-15)^2 + 100 - x^2 = 40 \Rightarrow x = ...$ | M1 | 3.1a |
| | Either $\Rightarrow -30x + 325 = 40 \Rightarrow x = 9.5$ Or $y = \dfrac{\sqrt{39}}{2} =$ awrt $\pm 3.12$ | A1 | 1.1b |
| | Attempts to find the angle $AOB$ in circle $C_1$ Eg Attempts $\cos\alpha = \dfrac{"9.5"}{10}$ to find $\alpha$ then $\times 2$ | M1 | 3.1a |
| | Angle $AOB = 2\times\arccos\left(\dfrac{9.5}{10}\right) = 0.635$ rads (3sf) * | A1* | 2.1 |
| | | (4) | |
| (b) | Attempts $10\times(2\pi - 0.635) = 56.48$ | M1 | 1.1b |
| | Attempts to find angle $AXB$ or $AXO$ in circle $C_2$ (see diagram) E.g. $\cos\beta = \dfrac{15 - "9.5"}{\sqrt{40}} \Rightarrow \beta = ...$ (Note $AXB = 1.03$ rads) | M1 | 3.1a |
| | Attempts $10\times(2\pi - 0.635) + \sqrt{40}\times(2\pi - 2\beta)$ | dM1 | 2.1 |
| | $= 89.7$ | A1 | 1.1b |
| | | (4) | |
| | | | (8 marks) |
| Notes: | | | |




**(a)**

**M1:** For the key step in an attempt to find either coordinate for where the two circles meet.
Look for an attempt to set up an equation in a single variable leading to a value for $x$ or $y$.

**A1:** $x = 9.5$ (or $y = \dfrac{\sqrt{39}}{2} =$ awrt $\pm 3.12$)

**M1:** Uses the radius of the circle and correct trigonometry in an attempt to find angle $AOB$ in circle $C_1$

E.g. Attempts $\cos\alpha = \frac{"9.5"}{10}$ to find $\alpha$ then $\times 2$

Alternatives include $\tan\alpha = \frac{\sqrt{100-"9.5"^2}}{"9.5"} = (0.3286...)$ to find $\alpha$ then $\times 2$

And $\cos AOB = \frac{10^2+10^2-\left(\sqrt{39}\right)^2}{2\times10\times10} = \frac{161}{200}$

**A1*:** Correct and careful work in proceeding to the given answer. Condone an answer with greater accuracy. Condone a solution where the intermediate value has been truncated, provided the trig equation is correct.

E.g. $\sin\alpha = \frac{\sqrt{39}}{20} \Rightarrow \alpha = 0.317 \Rightarrow AOB = 2\alpha = 0.635$

Condone a solution written down from awrt $36.4°$ (without the need to shown any calculation.)

E

**(b)**

**M1:** Attempts to use the formula $s = r\theta$ with $r = 10$ and $\theta = 2\pi - 0.635$

The formula may be embedded. You may see $\underline{\underline{2\pi 10}} + 2\pi\sqrt{40} \underline{\underline{-10\times0.635}}$... which is fine for this M1

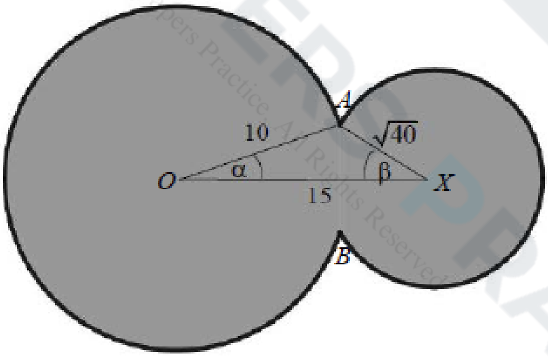
**M1:** Attempts to use a correct method in order to find angle $AXB$ or $AXO$ in circle $C_2$

Amongst many other methods are $\tan\beta = \frac{"3.12"}{15-9.5}$ and $\cos AXB = \frac{40+40-\left(\sqrt{39}\right)^2}{2\times\sqrt{40}\times\sqrt{40}} = \frac{41}{80}$

Note that many candidates believe this to be 0.635. This scores M0 dM0 A0

**dM1:** A full and complete attempt to find the perimeter of the region.
It is dependent upon having scored both M's.

**A1:** awrt 89.7




(a)

M1: For the key step in attempting to find all lengths in triangle $OAX$, condoning slips

A1: All three lengths correct

M1: Attempts cosine rule to find $\alpha$ then $\times 2$

A1*: Correct and careful work in proceeding to the given answer

Q13.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (a) | $t=0,\ \theta=18 \Rightarrow 18=A-B$ <br> **or** <br> $t=10,\ \theta=44 \Rightarrow 44=A-Be^{-0.7}$ | M1 | 3.1b |
| | $t=0,\ \theta=18 \Rightarrow 18=A-B$ <br> **and** <br> $t=10,\ \theta=44 \Rightarrow 44=A-Be^{-0.7}$ <br> **and** <br> $\Rightarrow A=\ldots, B=\ldots$ | M1 | 3.1a |
| | At least one of: $A=69.6$, $B=51.6$ but allow awrt 70/awrt 52 | A1 <br> **M1 on EPEN** | 1.1b |
| | $\theta=69.6-51.6e^{-0.07t}$ | A1 | 3.3 |
| | | **(4)** | |
| (b) | The maximum temperature is "69.6"(°C) (according to the model) <br> (The model has an) upper limit of "69.6"(°C) <br> (The model suggests that) the boiling point is "69.6"(°C) | B1ft | 3.4 |
| | Model is not appropriate as 69.6(°C) is much lower than 78(°C) | B1ft | 3.5a |
| | | **(2)** | |
| | | | **(6 marks)** |

**Notes:**

**(a)**

**M1:** Makes the first key step in the solution of the problem. Substitutes $t=0$ and $\theta=18$ **or** $t=10$ and $\theta=44$ into the equation of the model to obtain an equation connecting $A$ and $B$.
Note that $18=A-Be^0$ scores M0 unless $18=A-B$ is seen or implied later.
If they do not obtain an equation in $A$ and $B$ using the first conditions e.g. they have $18=A-1$ then they can score this mark if they substitute $A=19$ directly into $44=A-Be^{-0.7}$ as an equation in $A$ and $B$ is implied.

**M1:** Substitutes $t=0$ and $\theta=18$ **and** $t=10$ and $\theta=44$ to obtain 2 equations connecting $A$ and $B$ **and** then proceeds to solves their equations in $A$ and $B$ simultaneously to obtain values for both constants. Do not be too concerned with the processing as long as values for $A$ and $B$ are obtained.

**A1(M1 on EPEN):** For $A=$ awrt 70 **or** $B=$ awrt 52

**A1:** For $\theta=69.6-51.6e^{-0.07t}$ **Must be a <u>fully correct equation as shown</u> but allow recovery if seen in (b).**
Note that some candidates evaluate $e^0$ as 0 and so obtain $A=18$ and then write $44=18-Be^{-0.7}$ and solve for $B$.
Such attempts can score M1M0A0A0 only.

**(b)**

**B1ft:** Identifies $A$ as the boiling point/maximum temperature in the model. Follow through their $A$.

**B1ft:** Makes a valid conclusion (valid/not valid, good/not good etc.) that refers to the 78 and includes a reference to a significant/large difference

**Alternative** provided their $A < 78$

**B1ft:** $\theta=69.6-51.6e^{-0.07t}=78 \Rightarrow 51.6e^{-0.07t}=69.6-78=-8.4$

$\Rightarrow e^{-0.07t}=-\dfrac{7}{43}$ and $\ln\left(-\dfrac{7}{43}\right)$ and makes a reference to the fact that the equation cannot be solved or e.g. cannot take log of a negative number. You can condone numerical slips in the calculation.

**B1ft:** Model is not appropriate as 69.6(°C) is much lower than 78(°C)

**Minimum for both marks:** The model is not appropriate as "69.6"(°C) is much lower than 78(°C)

**Note that these marks are not available if their equation is solvable. Note also that B0B1 is not possible.**

Q14.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (i) | For setting up the contradiction:<br>There exists integers $p$ and $q$ such that $pq$ is even and both $p$ and $q$ are odd | B1 | 2.5 |
| | For example, sets $p=2m+1$ and $q=2n+1$ and then attempts $pq=(2m+1)(2n+1)=...$ | M1 | 1.1b |
| | Obtains $pq=(2m+1)(2n+1)=4mn+2m+2n+1$<br>$=2(2mn+m+n)+1$<br>States that this is odd, giving a contradiction so<br>" if $pq$ is even, then at least one of $p$ and $q$ is even" * | A1* | 2.1 |
| | | (3) | |
| (ii) | | | |
| | $(x+y)^2<9x^2+y^2 \Rightarrow 2xy<8x^2$ | M1 | 2.2a |
| | States that as<br>$x<0 \Rightarrow 2y>8x$<br>$\Rightarrow y>4x$ * | A1* | 2.1 |
| | | (2) | |
| | | | (5 marks) |
| Notes: | | | |

(i)

B1: For using the "correct"/ allowable language in setting up the contradiction.

Expect to see a minimum of

- "assume" or "let" or "there is " or other similar words
- "$pq$ is even" and "$p$ and $q$ are (both) odd"

M1: Uses a correct algebraic form for $p$ and $q$ and attempting to multiply.

Allow any correct form so $p=2n+1$ and $q=2m+3$ would be fine to use

**Different variables must be used** for $p$ and $q$, so $p=2n+1$ and $q=2n-1$ would be M0

A1*: Full argument .

This requires (1) a correct calculation for their $pq$

(2) a correct reason and conclusion that it is odd

E.g. $(2m+1)(2n+1)=4mn+2m+2n+1=2(2mn+m+n)+1=\text{odd}$

E.g. $(2m-1)(2n+1)=4mn+2m-2n-1=\text{even}+\text{even}-\text{even}-1=\text{odd}$

and (3) a minimal statement implying that they have proven what was required which could be QED, proven etc

Note that B0 M1 A1 is possible

(ii)

M1: For multiplying out and cancelling terms before proceeding to a correct intermediate line such as $2xy < 8x^2$ o.e. such as $2x(4x-y) > 0$

A1*: Full and rigorous proof with reason shown as to why inequality reverses. The point at which it reverses must be correct and a correct reason given

See scheme

Alt: $2xy < 8x^2 \Rightarrow xy - 4x^2 < 0 \Rightarrow x(y-4x) < 0$

as $x < 0$, $(y-4x) > 0 \Rightarrow y > 4x$ scores M1 A1

So, the following should be scored M1 A0 as line 3 is incorrect

$2xy - 8x^2 < 0$

$\Rightarrow 2xy < 8x^2$

$\Rightarrow y < 4x$

$\Rightarrow y > 4x$ as $x < 0$

There should be no incorrect lines in their proof